Content

Learning objectives and content

Objectives
The main focus of this course is on the theoretical and statistical foundations of Machine Learning with particular emphasis on modelling assumptions, estimation and evaluation. A secondary focus is on low-level practical aspects (e.g. vanilla implementations of various models and algorithms). After completing the course, students will be able to:

* analyze and assess the suitability of common machine learning methods for a given learning problem based on their underlying assumptions and statistical properties of the data
* compare weaknesses and strengths of common machine learning methods in terms of bias-variance trade-offs, generalization and robustness
* implement some of the most widely used methods and tackle limitations of naive implementations, such as numerical issues arising in practice
* critically reflect on model choice, theoretical guarantees and the impact of chosen evaluation metrics on empirical results

Content
* Introduction to machine learning and optimization; Instructions: Python for Data Mining and Machine Learning; Recap mathematical basics
* Regression (linear and non-linear): regularization, bias-variance trade-off, Ridge regression, Lasso
* Classification: Naive Bayes, k-Nearest Neighbors, Support vector machines and the kernel trick, Decision Trees and Random Forests; Evaluation of classifiers
* Neural Networks (Multi-layer Perceptrons and Convolutional Neural Networks), Backpropagation, Stochastic Gradient Descent, training techniques
* Unsupervised Learning: PCA, SVD, Recommender systems, K-Means clustering, Kernel k-means and Spectral Clustering

The course content is detailed in the accompanying Jupter book

2026

Statistics and Machine Learning (2IRR50)

Practical information

Course module
2IRR50

Course type
Bachelor College

Credits (ECTS)
5 EC

Category
Advanced

Language of instruction
English

Offered by
Mathematics and Computer Science - Computer Science domain

Remark
start in 2024

Course enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Course.

Enrolment periods

- Block 4
  Timeslot(s)
  A - Mo 1-4, We 9-10, Th 5-8

  Enrolment period
  15 November 2026 until 21 March 2027 23:59

  Unenrolment
  until 19 April 2027 23:59

Entrance requirements

Assumed previous knowledge
While there are no strict requirements, the course assumes that the students have followed the following courses: 2WCB0 (Calculus); 2DI90 (Probability and statistics); Data Analytics for Engineers (2IAB0); 2DRR00 : Linear Algebra and Applications. We further assume that the students have learned to program in Python.

Entrance requirements course enrolment
You must meet the following requirements

- Completed none of the course modules listed below
  - Data Mining and Machine Learning (2IIG0)

Instructional modes

Instructional modes

- Instruction
  Remark
- Lecture
  Remark

Tests

Tests

- Assignment 1
  Interim examination

  Test weight
  20

  Minimum grade
- Assignment 2
  Interim examination

  Test weight
  20

  Minimum grade
- Schoolyear ANS
  Final examination

  Test weight

60

Minimum grade
5.0

Test enrolment

OSIRIS Student
You can register yourself via OSIRIS Student. To do so, go to Enroll / Test.

Assignment 1

- Block 4
  Enrolment period
  15 November 2026 until 6 June 2027 23:59

Assignment 2

- Block 4
  Enrolment period
  15 November 2026 until 6 June 2027 23:59

Schoolyear ANS

- Block 4
  Enrolment period
  15 November 2026 until 6 June 2027 23:59

  Unenrolment
  until 6 June 2027

  Test date
  21 June 2027

- Block I
  Enrolment period
  15 November 2026 until 1 August 2027 23:59

  Unenrolment
  until 1 August 2027

  Test date
  16 August 2027

Lecturer(s)

Contactperson for the course

- dr. S.C. Hess

Responsible lecturer

- dr. S.C. Hess

Subject matter expert

- dr. S.C. Hess

Materials

Do you want to know more?

dr. S.C. Hess can tell you more about this course.
Send an e-mail to s.c.hess@tue.nl